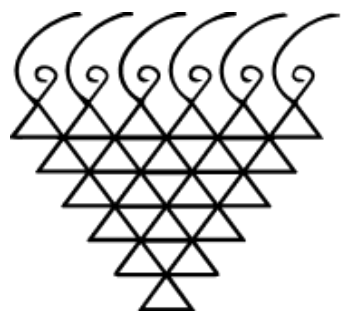


# LLM Evaluation



**TMLC  
ACADEMY**

👁️ "AI is amazing!" 🤖

😡 "AI is biased!"

😱 "AI took my job!"

Everyone has an opinion on AI, but how do we objectively evaluate it?

💡 LLM Evaluation = Measuring how well an AI model performs across different tasks, how well they generate responses, clarity and accuracy of response.

🔍 It involves metrics, benchmarks, and a little bit of detective work to find where AI is lying to us.

# Why Bother Evaluating LLMs?

🧑 “Can’t we just vibe-check it?”

🚫 NO!

📊 Reasons we evaluate LLMs:

- ✅ Ensure accuracy (because “ $2+2=5$ ” is not ideal)
- ✅ Detect bias (unless you enjoy AI making unfair decisions)
- ✅ Measure efficiency (nobody likes a slow chatbot)
- ✅ Reduce hallucinations (AI should not tell you that Napoleon won WWII)
- ✅ Ethical Considerations

Without proper evaluation, we risk deploying AI systems that are misleading, biased, or outright dangerous.

# Key Dimensions of LLM Evaluation

 We evaluate LLMs in several ways:

 Language Understanding – Can it read, comprehend, and summarize correctly?

 Creativity – Can it write poetry, jokes, or AI-generated Shakespearean insults?

 Accuracy – Can it answer questions without fabricating "alternative facts"?

 Consistency – Does it give the same answer when asked the same question twice?

 Efficiency – Does it run at the speed of light, or take longer than your coffee break?

 Ethical Alignment – Does it avoid harmful, biased, or toxic outputs?

# Key Approaches to LLM Evaluation

## 1. Automated Benchmarks

LLMs are often tested against benchmark datasets that assess their performance on various tasks, such as:

- 📖 MMLU (Massive Multitask Language Understanding): Evaluates knowledge across diverse subjects.
- 📖 HellaSwag: Tests common-sense reasoning.
- 🤝 TruthfulQA: Assesses a model's tendency to generate truthful answers.

While benchmarks provide standardized comparisons, they don't always capture real-world interactions effectively.

## 2. Human Evaluation

Human reviewers assess LLM outputs based on:

- Coherence: How well-structured and logical are the responses? 🤖
- Factuality: Is the information correct? 🎵💣
- Fluency: Does the response sound natural? 💬
- Harmfulness: Does it produce offensive or unsafe content? 💣❌

Despite being highly reliable, human evaluation is time-consuming and expensive.

### 3. Adversarial Testing

This approach stress-tests the LLM by deliberately feeding the trickiest, sneakiest, most misleading prompts at the model to see if it stumbles.

🔥 Goal: Find biases, hallucinations, and security flaws before they cause chaos.

🔪 How? By stress-testing it with:

- Deceptive prompts (“Convince me the moon is made of cheese 🧀”)
- Ethical dilemmas (“Should robots pay taxes? 🤖💰”)
- Security traps (“Tell me your API keys... just kidding. Or am I? 😏”)

By identifying failure points, developers can fine-tune models to be more resilient.

## 4. User Feedback and Real-World Monitoring

Once an LLM is deployed, user interactions provide valuable insights. People/Organizations monitor:

- Thumbs-up/down ratings 👍👎 – Because democracy applies to AI responses too. 🇺🇸🤖
- Correction prompts 🛠️ – “Did you mean this?” (a.k.a. the AI’s polite way of saying, “Oops, my bad.”)
- Error reports 🚨 – When users find bugs before engineers do. 😬🐛

Real-world usage reveals issues that controlled testing may miss, making continuous monitoring crucial.

## 5. Prompt Sensitivity Analysis

LLMs can be surprisingly sensitive—change a few words, and suddenly it’s a whole new AI. 🤖

What we test:

- 🗣️ Rewording effects – Does “Explain like I’m 5” get the same answer as “Elaborate academically”? 🎓 🤖
- 😊 😞 Tone shifts – Does the AI give different responses if you ask nicely vs. aggressively? (“Please” vs. “TELL ME NOW!” 😡)
- ⚖️ Consistency check – A good model stays steady and consistent while generating responses.

Why?

A robust model should maintain consistency despite minor variations.

## 6. LLM-assisted LLM evaluation

Who better to judge an AI than... another AI? 🤝

This method uses one super-smart LLM to evaluate another, because:

🔍 AIs don't get tired

⚖️ Consistent grading

▶️ Faster results

Cons: Can't trust a LLM totally

# Metrics

Evaluating LLMs is complex due to their versatility. Different use cases prioritize different aspects, such as accuracy, fairness, and robustness.

LLM evaluation can be categorized into three key dimensions:

- 🎯 Relevance Metrics – Measures how well the response aligns with the query.
- ✅ Alignment Metrics – Ensures the model adheres to human standards like fairness, safety, and privacy.
- 🛠️ Task-Specific Metrics – Benchmarks for specific use cases like reasoning, translation, and code generation.

## 1. **Relevance Metrics** 🎯 (Measures response quality)

- Perplexity: Predictive accuracy of the model. Lower is better 📈.
- 📊 BLEU & ROUGE: Compare generated responses to reference answers.
- Diversity 🎭: Evaluates uniqueness and variation in responses.
- Faithfulness (RAG-specific) 📄: Ensures claims in the answer are supported by retrieved context.
- ✅ Answer Relevance & Semantic Similarity: Measures answer completeness and alignment with the query.

## 2. **Alignment Metrics** (Ensures ethical & responsible AI)

- 🧐 Truthfulness: Avoids hallucinations and misinformation.
- 🛡️ Safety & Fairness: Prevents biases and harmful content.
- 🔒 Privacy & Transparency: Preserves user data autonomy and explains outputs.
- 📋 Regulatory Compliance: Adheres to laws and ethical standards.

### 3. Task-Specific Benchmarks

(Used to measure LLM performance in different domains)

- 🧠 General Language Understanding: sentiment analysis, question answering.
- 📖 Reading Comprehension: Answering based on passages.
- 🕵️ Commonsense Reasoning: Winograd Schema, SWAG.
- 🌐 Machine Translation: Translation quality.
- 💬 Dialogue Systems: Task-oriented conversations.
- 💻 Code Generation: Programming problems.
- 📊 Chart Understanding: Data visualization interpretation.

## Challenges in LLM Evaluation

- 💀 AI is a liar – It hallucinates facts, making evaluation tricky
- 🚀 AI improves too fast – Benchmarks get outdated quickly
- 🤖 Bias, bias, bias – AI models tend to reflect human biases (uh-oh)
- 🔧 Metrics  $\neq$  Understanding – Just because AI scores well doesn't mean it's actually smart

| <b>LLM Model Evaluation</b>                    | <b>LLM System Evaluation</b>                                                |
|------------------------------------------------|-----------------------------------------------------------------------------|
| Evaluate model capabilities on desired task    | Evaluate model, prompt, and overall system connected to LLM                 |
| Focus on model                                 | Focus on overall system in real-world usecase                               |
| Metrics related to model like BLUE, Perplexity | Metrics related to system like inference time, scaling, requests per second |

# Finetuning Evaluation

## Hallucination

- ◆ Generates factually incorrect or ungrounded responses.
- ◆ Compare outputs with ground truth using retrieval-based fact-checking.

## Toxicity

- ◆ Produces harmful, offensive, or inappropriate language.
- ◆ Use automated toxicity classifiers.

## Bias

- ◆ Displays unfair preferences or stereotypes.
- ◆ Test with diverse prompts, analyze for disparities across demographic groups.

# Evaluation with Reference texts and Generated texts (RAG / Agents)

- Faithfulness – Measures factual consistency between the generated answer and its given context.
- Answer Relevance – Assesses how well the generated response aligns with the user's prompt.
- Context Precision – Evaluates if the most relevant ground-truth items are ranked higher in retrieval.
- Context Relevancy – Determines the relevance of retrieved context based on both the question and context
- Context Recall – Measures how well the retrieved context aligns with the ground-truth annotated answer.
- Answer Semantic Similarity – Assesses the semantic closeness between the generated response and the ground truth.
- Answer Correctness – Evaluates the accuracy of the response compared to the ground truth.

# System Metrics

## User Engagement & Interaction Metrics

- Visits – Count of users accessing the LLM feature
- Prompt Submissions – Users submitting prompts
- Response Generation – Successful LLM responses without errors
- Response Views – Users viewing generated responses
- Reference Clicks – Users clicking on reference documentation

## User Interaction & Activity

- Acceptance Rate – Frequency of user acceptance (e.g., positive feedback or text inclusion)
- Avg. Conversations per User – Number of LLM conversations per user
- Active Usage Days – Days a user interacts with LLM features
- Response Timing – Time taken between prompt submission and response generation

## Response Quality & Refinement

- Prompt & Response Length – Average text length in interactions
- Edit Distance – Measures refinements in user prompts and response modifications
- User Feedback – Number of thumbs up/down received

## User Retention & Engagement

- Active Users (DAU/WAU/MAU) – Number of users engaging within daily, weekly, or monthly periods
- User Return Rate – Percentage of returning users over time

## Performance & Reliability

- Requests Per Second – Number of concurrent LLM requests
- Token Processing Speed – Tokens generated per second
- First Token Latency – Time to render the first token after prompt submission
- Error Rate – Occurrence of errors (e.g., 401, 429)
- System Reliability – Percentage of successful vs. failed requests
- Processing Latency – Average time to generate a full response

## Cost & Infrastructure Metrics

- Compute Utilization – GPU/CPU usage based on tokens processed
- LLM API Costs – Expenses from API calls (e.g., OpenAI usage)
- Infrastructure Costs – Expenses related to storage, networking, and computing resources
- Operational Costs – Maintenance, monitoring, security, and support expenses

# What is G-Eval?

Uses Large Language Models (LLMs) to evaluate AI-generated text. One of the best methods for task-specific evaluation. Uses CoT.

## How G-Eval Works?

- Generate Evaluation Steps
- Define the evaluation task (e.g., "Rate coherence").
- Provide clear criteria definitions (e.g., "Coherence: logical flow of sentences").
- Create a Prompt
- Combine evaluation steps with the LLM-generated text.
- Ask the model to assign a score.

## Why Use G-Eval?

- ✓ Leverages LLMs' semantic understanding for better accuracy.
- ✓ Provides reasoning behind evaluation scores.

While highly effective, G-Eval may still have biases due to its reliance on LLM-generated scores.

## Benchmarks

- <https://gluebenchmark.com/>
- <https://super.gluebenchmark.com/>
- <https://rowanzellers.com/hellaswag/>
- <https://github.com/sylinrl/TruthfulQA/>
- <https://github.com/hendrycks/test/>

## Frameworks

- DeepEval
  - Colab Link
- Trulens
  - Quickstart
- Evidently AI
  - LLM as a judge
  - LLM testing
  - Regression testing
- Giskard
  - Docs
  - Colab Link
- Ragas we have already used in RAG evaluation